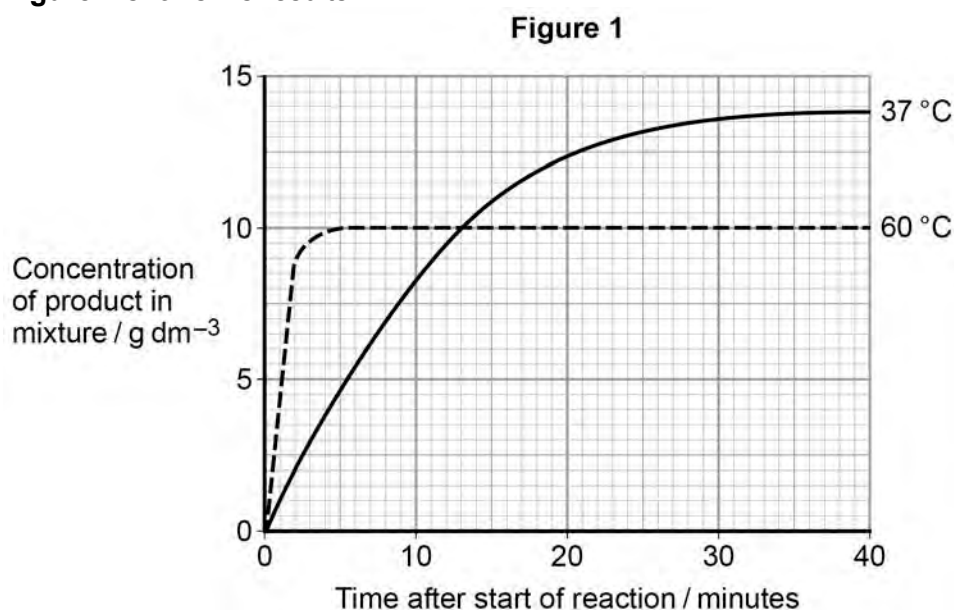


1

A technician investigated the effect of temperature on the rate of an enzyme-controlled reaction. At each temperature, he started the reaction using the same concentration of substrate.

Figure 1 shows his results.



0 1 . 1 Give **two** other factors the technician would have controlled.

[1 mark]

1 _____

2 _____

0 1 . 2 Draw a tangent on each curve to find the initial rates of reaction. Use these values to calculate the ratio of the initial rates of reaction at $60^\circ\text{C} : 37^\circ\text{C}$. Show your working.

[2 marks]

Ratio = _____ :1

0	1
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 .

3

Explain the difference in the initial rate of reaction at 60 °C and 37 °C.

[2 marks]

0	1
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 .

4

Explain the difference in the rates of reaction at 60 °C and 37 °C between 20 and 40 minutes.

[4 marks]

[Extra space]
